

Survival Analysis: Hypothesis Testing

Lectures 13 & 14 — Cheat Sheet

BIOST 3050 / STAT 2261 | Spring 2026 | Prof. Ying Ding

Overview: Three Test Settings

- One-sample** Does data fit a known model $S_0(t)$?
Two-sample Do 2 groups share the same survival?
K-sample Do $K \geq 2$ groups share the same survival?

All hypotheses can be equivalently stated as:

- $H_0 : S(t) = S_0(t)$, or $H(t) = H_0(t)$, or $h(t) = h_0(t)$

Key challenge: Right censoring and left truncation prevent standard tests (e.g. t -test, ANOVA).

One-Sample Test

Data: $\{(T_i, \delta_i), i = 1, \dots, n\}$ with $T_i = \min(X_i, C_i)$, $\delta_i = I(T_i = X_i)$.

Null: $H_0 : h(t) = h_0(t)$ for all $t \in [0, \tau]$.

Notation: $t_1 < \dots < t_D$: unique death times; d_j : deaths at t_j ; $Y_j = Y(t_j)$: at risk at t_j .

General Test Statistic

With weight function $W(t)$:

$$Z(\tau) = \underbrace{\sum_{j=1}^D W(t_j) \frac{d_j}{Y(t_j)}}_{\text{Observed}} - \underbrace{\int_0^\tau W(s) h_0(s) ds}_{\text{Expected}}$$

Variance under H_0 : $V[Z(\tau)] = \int_0^\tau \frac{W^2(s) h_0(s)}{Y(s)} ds$

Test: $Z(\tau)^2 / V[Z(\tau)] \sim \chi_1^2$ under H_0 .

Log-Rank Test ($W(t) = Y(t)$)

The most popular choice. Simplifications:

No left truncation ($L_i = 0$):

$$O(\tau) = \sum_{i=1}^n \delta_i \quad (\text{total events})$$

$$E(\tau) = \sum_{i=1}^n H_0(T_i) \quad (\text{expected events})$$

$$V[Z(\tau)] = E(\tau) = \sum_{i=1}^n H_0(T_i)$$

Key property: $E(\tau) = V[Z(\tau)]$ for the one-sample log-rank. The test stat becomes $T_1 = (O - E) / \sqrt{E}$.

With left truncation (delayed entry L_i):

$$E(\tau) = V[Z(\tau)] = \sum_{i=1}^n [H_0(T_i) - H_0(L_i)]$$

One-Sided Alternatives

- $H_A : h(t) > h_0(t)$: reject for large positive T_1
- $H_A : h(t) < h_0(t)$: reject for large negative T_1

Harrington–Fleming Weights

$$W(t) = Y(t) [S_0(t)]^p [1 - S_0(t)]^q, \quad p, q \geq 0$$

Choice	Sensitivity
$p > q$	Early departures from H_0
$q > p$	Late departures from H_0
$p = q$	Mid-range departures
$p = q = 0$	Standard log-rank

IBCSG Trial 6 Example

Breast cancer, $n = 735$, $r = 210$ deaths, $\sum T_i = 46,296$ months.

$\hat{\lambda} = 210 / 46,296 \approx 0.0045$ (MLE for Exponential).

H_0	T_1	p -value
$H_0(t) = 0.0045t$	0.116	0.27
$H_0(t) = 0.004t$	1.8	0.07
$H_0(t) = 0.005t$	-1.4	0.16
$H_0(t) = 0.006t$	-4.1	< 0.01

R: `survdifff(Surv(T,delta) ~ offset(sp))` where `sp = exp(-0.0045*T)`.

K-Sample Tests

Hypotheses: $H_0 : h_1(t) = h_2(t) = \dots = h_K(t)$ for all $t \leq \tau$.

$\tau = \min_k t_{k,\max}$ (all groups must have ≥ 1 at risk).

Pooled data notation: d_j = total deaths at t_j ; d_{jk} = deaths in group k ; Y_j = total at risk; Y_{jk} = at risk in group k .

Test Statistic for Group k

With common weight $W(t_j)$ and $W_k(t_j) = Y_{jk} W(t_j)$:

$$Z_k(\tau) = \sum_{j=1}^D W(t_j) \left[d_{jk} - d_j \frac{Y_{jk}}{Y_j} \right]$$

Interpretation: Z_k = Observed – Expected deaths in group k (weighted).

Variance & Covariance

$$\hat{\sigma}_{kk} = \sum_j W(t_j)^2 \frac{Y_{jk}}{Y_j} \left(1 - \frac{Y_{jk}}{Y_j}\right) \frac{Y_j - d_j}{Y_j - 1} d_j$$

$$\hat{\sigma}_{kk'} = - \sum_j W(t_j)^2 \frac{Y_{jk} Y_{jk'}}{Y_j^2} \frac{Y_j - d_j}{Y_j - 1} d_j$$

Global Test Statistic

Key fact: $\sum_{k=1}^K Z_k(\tau) = 0 \Rightarrow$ covariance matrix Σ is singular. Drop the last group!

$$\chi^2 = (Z_1, \dots, Z_{K-1}) \hat{\Sigma}_{(K-1) \times (K-1)}^{-1} (Z_1, \dots, Z_{K-1})^T$$

$$\chi^2 \sim \chi_{K-1}^2 \quad \text{under } H_0$$

Two-Sample Test ($K = 2$)

The most common clinical trial scenario.

Log-Rank ($W(t_j) = 1$)

$$Z_1(\tau) = \sum_{j=1}^D \left[d_{j1} - \frac{Y_{j1}}{Y_j} d_j \right]$$

$$\widehat{\text{Var}}(Z_1) = \sum_{j=1}^D \frac{Y_{j1}}{Y_j} \left(1 - \frac{Y_{j1}}{Y_j}\right) \frac{Y_j - d_j}{Y_j - 1} d_j$$

If **no ties** ($d_j = 1$): $\widehat{\text{Var}}(Z_1) = \sum_j \frac{Y_{j1}}{Y_j} \left(1 - \frac{Y_{j1}}{Y_j}\right)$

Standardized Statistic

$$Z = \frac{Z_1(\tau)}{\sqrt{\widehat{\text{Var}}(Z_1(\tau))}} \stackrel{H_0}{\sim} N(0, 1)$$

Equivalence: $Z^2 = \chi_1^2$ (same as K -sample with $K = 2$).

Decision Rules

Test	Stat	Reject H_0 if
Two-sided	Z	$ Z > z_{1-\alpha/2}$
One-sided	Z	$Z > z_{1-\alpha}$ or $Z < -z_{1-\alpha}$
$K > 2$	χ^2	$\chi^2 > \chi_{\alpha, K-1}^2$

One-sided tests are **only** available for $K = 2$. For $K > 2$, you **must** use χ^2 (two-sided).

Weight Functions for K -Sample

Name	$W(t_j)$	Best for
Log-Rank	1	Proportional hazards
Gehan	$Y(t_j)$	Early differences
Tarone-Ware	$\sqrt{Y(t_j)}$	Compromise
Fleming-H.	$[\hat{S}(t_{j-1})]^p [1 - \hat{S}(t_{j-1})]^q$	Flexible

Crossing curves? Log-rank may have low power. Use Gehan (early) or F-H with $q > 0$ (late).

Kidney Dialysis Example

Surgical (grp 1) vs. percutaneous (grp 2) catheter placement.

$$\sum d_{j1} = 15, \sum E_{j1} = 11.036, Z_1 = 3.964$$

$$\widehat{\text{Var}}(Z_1) = 6.211$$

$$Z = \frac{3.964}{\sqrt{6.211}} \approx 1.59 \quad (p > 0.05)$$

Result: Not significant at $\alpha = 0.05$. Curves cross: surgical better early, percutaneous better late. Different weights $\Rightarrow$ potentially different conclusions.

R Implementation

- `survdif(Surv(time,status) ~ group)`
- `survdif(..., rho=0) → Log-rank`
- `survdif(..., rho=1) → Modified Peto-Peto`

Master Formula Summary

One-Sample Log-Rank	
O	$\sum_i \delta_i$
$E = V$	$\sum_i H_0(T_i) \quad [\text{or } \sum_i [H_0(T_i) - H_0(L_i)]]$
Test	$T_1 = (O - E) / \sqrt{E} \sim N(0, 1)$
Two-Sample / K -Sample	
Z_k	$\sum_j W_j [d_{jk} - d_j Y_{jk} / Y_j]$
Constraint	$\sum_k Z_k = 0$ (drop one group)
$K = 2$	$Z = Z_1 / \sqrt{\widehat{\text{Var}}(Z_1)} \sim N(0, 1)$
$K > 2$	$\chi^2 = \mathbf{Z}^T \hat{\Sigma}^{-1} \mathbf{Z} \sim \chi_{K-1}^2$